GAMIFICATION TECHNIQUES

Derived from

You Are Not So Smart

by David McRaney

12 Psychological Principles Applied to Game Design

About This Document

This draft identifies 12 gamification techniques directly applicable to a psychology-based game built around the concepts explored in David McRaney's book. Each technique maps a cognitive bias, heuristic, or logical fallacy from the book onto a concrete game mechanic, creating challenges where players confront and navigate their own mental blind spots.

Each card below outlines: the original misconception and truth from the book, how it translates into a gameplay challenge, and the specific mechanic that operationalizes it.

01 Priming | Chapter: Ch. 1

📌 **Misconception:** *You know when you are being influenced and how it affects your behavior.*

✅ **Truth:** *Unconscious cues constantly shape your decisions and actions without your awareness.*

🎮 **Game Application:** Before presenting a puzzle or challenge, subtly expose players to thematic cues — colors, images, words — that prime them toward a desired cognitive state or strategy. A mystery level could use dark visuals and ominous ambient sound to prime cautious thinking.

⚙️ **Mechanic:** Environmental Priming — theme each level's intro screen, music, and color palette to unconsciously guide player mindset and decision style before gameplay begins.

02 Confirmation Bias | Chapter: Ch. 3

📌 **Misconception:** *Your opinions are the result of years of rational, objective analysis.*

✅ **Truth:** *You notice and remember information that confirms what you already believe, and ignore the rest.*

🎮 **Game Application:** Present players with ambiguous clues or red herrings and let them form a hypothesis. Their confirmation bias will make them collect evidence supporting that hypothesis and overlook contradicting clues — teaching them to challenge their own assumptions to progress.

⚙️ **Mechanic:** The False Trail Puzzle — players form a working theory early; the game rewards those who revisit and question their assumptions before locking in their answer.

03 The Dunning-Kruger Effect | Chapter: Ch. 11

📌 **Misconception:** *You can predict how well you would perform in any situation.*

✅ **Truth:** *You are generally poor at estimating your own competence and the difficulty of complex tasks.*

🎮 **Game Application:** Let players self-assess their skill level before each challenge. Track how their prediction compares to actual performance. Players who are overconfident face harder follow-up challenges; those who underestimate unlock a hidden 'Humble Expert' bonus track.

⚙ **Mechanic:** Confidence Calibration — a pre-round self-rating slider. Post-round, show a 'confidence vs. reality' score to build metacognitive awareness over time.

04 Availability Heuristic | Chapter: Ch. 9

📌 **Misconception:** *You understand the world based on statistics and facts from many examples.*

✅ **Truth:** *You believe something is more common if you can easily think of one vivid example, regardless of actual frequency.*

🎮 **Game Application:** Present players with sensational-sounding but rare scenarios vs. mundane but common ones. Players must judge which is statistically more likely. Trick questions exploit the availability heuristic — players who think critically about data rather than vivid recall succeed.

⚙ **Mechanic:** Probability Duel — two competing facts are shown; one sounds dramatic, one sounds boring. Players bet on which is more statistically true, learning to separate emotional salience from actual frequency.

05 Hindsight Bias | Chapter: Ch. 4

📌 **Misconception:** *After you learn something new, you remember how you were once ignorant or wrong.*

✅ **Truth:** *You often look back on things you just learned and falsely assume you knew them all along.*

🎮 **Game Application:** Show players a complex scenario and ask them to predict the outcome. After the reveal, ask them to recall their original prediction. Their tendency to misremember their prediction as 'correct all along' is the challenge — the game compares their written-down answer to what they claim they said.

⚙ **Mechanic:** The Memory Vault — players write their prediction in a locked journal before each scenario resolves. Afterward, they see their actual entry versus what they claimed to remember, scoring points for honest self-recall.

06 The Bystander Effect | Chapter: Ch. 10

📌 **Misconception:** *When someone is hurt, people rush to their aid.*

✅ **Truth:** *The more people present, the less likely any single person is to help — responsibility gets diluted.*

🎮 **Game Application:** In multiplayer rounds, create a 'help needed' event visible to all players. The larger the group that notices, the less likely spontaneous action is. Points go to the first player who breaks from the group and acts — teaching players that action requires individual initiative, not crowd following.

⚙️ **Mechanic:** The Diffusion Challenge — a group event where everyone can see a problem. Individual XP rewards decay the longer the group waits collectively, incentivizing the first actor.

07 The Anchoring Effect | Chapter: Ch. 39

📌 **Misconception:** *You evaluate information objectively on its own merits.*

✅ **Truth:** *The first piece of information you encounter heavily influences all subsequent judgments.*

🎮 **Game Application:** Present players with a misleading 'anchor' number or value before asking them to estimate or choose. Players who recognize the anchor and adjust their reasoning appropriately score higher than those who let the initial number dominate their answer.

⚙️ **Mechanic:** The Anchor Round — players see a large, obviously wrong number first (e.g., 'Is the answer more or less than 10,000?'), then must give their real estimate. Bonus points for landing within a narrow correct range despite the anchor.

08 Procrastination & Present Bias | Chapter: Ch. 6

📌 **Misconception:** *You procrastinate because you are lazy and cannot manage your time.*

✅ **Truth:** *Procrastination is driven by present bias — you systematically overvalue immediate rewards and undervalue future ones.*

🎮 **Game Application:** Give players a resource they can spend now for a small reward or save for a compounding future reward. Players who consistently choose delayed gratification unlock powerful late-game tools. A countdown timer that shrinks rewards the longer you wait operationalizes hyperbolic discounting.

⚙️ **Mechanic:** The Marshmallow Bank — players deposit earned tokens at the start of each round. Tokens held for 3 rounds triple in value. Players must resist spending for short-term gains to maximize long-term scores.

09 Apophenia (Pattern Seeking) | Chapter: Ch. 12

📌 **Misconception:** *Miraculous coincidences must have hidden meaning.*

✅ **Truth:** *Coincidences are routine; any meaning applied to them comes from your own mind, not from the data.*

🎮 **Game Application:** Scatter random data patterns across the game world with a few real patterns hidden among them. Players who chase every coincidence waste time; those who apply statistical thinking to distinguish signal from noise find the true clues faster.

Mechanic: Signal vs. Noise Board — players mark which patterns they believe are meaningful. They earn points for correctly identifying real patterns and lose points for false positives, training them to resist seeing meaning in randomness.

10 Normalcy Bias | Chapter: Ch. 7

Misconception: *Your fight-or-flight instincts kick in and you panic when disaster strikes.*

Truth: *You often become abnormally calm and pretend everything is normal in a crisis, delaying action.*

Game Application: Introduce slow-building crises that escalate over multiple rounds without obvious alarm signals. Players who ignore early warning signs (because the situation seems 'not that bad yet') are overtaken by the scenario. Players who act early on subtle cues — like real disaster responders — score highest.

Mechanic: The Slow Burn Scenario — a compound problem grows each round. Early interventions cost few resources; late interventions are expensive or impossible. Players learn to recognize and overcome their own normalcy bias.

11 The Illusion of Control | Chapter: Ch. 47

Misconception: *You have meaningful influence over outcomes that are actually random.*

Truth: *You frequently believe your actions cause results in situations governed purely by chance.*

Game Application: Include rounds with hidden random outcomes. Players make choices that feel meaningful but secretly have no effect on results. After the round, reveal which decisions mattered and which were illusions. Players learn to distinguish skill from luck.

Mechanic: The Fake Lever — some buttons or choices in a round secretly do nothing. Post-round analytics show which decisions had causal impact. Players who correctly identify the 'real' controls across multiple rounds earn a 'Control Myth Buster' badge.

12 The Texas Sharpshooter Fallacy | Chapter: Ch. 5

Misconception: *You take randomness into account when determining cause and effect.*

Truth: *You ignore random chance when results seem meaningful, drawing conclusions from clusters that are just coincidence.*

Game Application: Show players a scatter of historical outcomes and ask them to identify the 'cause.' Some clusters are genuine, some are random. Players must resist drawing bull's-eyes around random clusters and instead use real statistical tests to identify true patterns.

Mechanic: The Barn Wall — players are shown data points and must place a 'cause marker' only where they have strong evidence. Markers placed on random clusters are penalized; accurate markers on genuine patterns score maximum points.

Quick Reference Summary

#	Technique	Core Mechanic
01	Priming	Environmental Priming
02	Confirmation Bias	The False Trail Puzzle
03	The Dunning-Kruger Effect	Confidence Calibration
04	Availability Heuristic	Probability Duel
05	Hindsight Bias	The Memory Vault
06	The Bystander Effect	The Diffusion Challenge
07	The Anchoring Effect	The Anchor Round
08	Procrastination & Present Bias	The Marshmallow Bank
09	Apophenia (Pattern Seeking)	Signal vs. Noise Board
10	Normalcy Bias	The Slow Burn Scenario
11	The Illusion of Control	The Fake Lever
12	The Texas Sharpshooter Fallacy	The Barn Wall

Based on You Are Not So Smart by David McRaney (2011, Dutton/Penguin)